

Generation Interconnection System Impact Study Report

For Queue Project Q13-0233

Sony 138 kV

May 02, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0233. The Interconnection Customer proposes a 447.2 MW wind facility interconnecting at the Sony 138 kV substation in the Maryland area. The request was received on April 26, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 141 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,499,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Seek Lane – Bear Swamp 138.0 kV: 194.7% loading (176 MVA) following the outage of the Sony – Woodenshoe 138.0 kV L; DFAX = 100.0%.

Bear Swamp – Wheaton 138.0 kV line: 185.2% loading (200 MVA) following the outage of the Sony – Woodenshoe 138.0 kV; DFAX = 100.0%.

Sony – Wheaton 138.0 kV line: 184.7% loading (217 MVA) following the outage of the Sony – Woodenshoe Substation 138.0 kV; DFAX = 100.0%.

The Woodenshoe – Sony 138.0 kV is overloaded to 172.3% of its 233 MVA emergency rating for the loss of the Sony – Wheaton 138.0 kV. The project contributes 100.0% DFAX (447.2 MW) to this overload.

For the loss of Tramway – Rossmoor South 138.0 kV, the Sligo – Linden 138.0 kV loads to 142.0% of its 179 MVA rating. Project DFAX: 12.7%.

Kensington – Woodenshoe 138.0 kV L: 124.7% loading (225 MVA) following the outage of the Wheaton – Sony 138.0 kV Line; DFAX = 61.5%.

For the loss of Grant Avenue – Seek Lane 138.0 kV, the Tokoma – Seek Lane 138.0 kV loads to 121.6% of its 194 MVA rating. Project DFAX: 21.6%.

The Darien – Sligo 138.0 kV is overloaded to 117.8% of its 193 MVA emergency rating for the loss of the Tramway – Rossmoor S 138.0 kV. The project contributes 12.7% DFAX (56.8 MW) to this overload.

For the loss of Tokoma – Seek Lane 138.0 kV Line, the Grant Avenue – Seek Lane 138.0 kV loads to 113.9% of its 233 MVA rating. Project DFAX: 20.9%.

Kensington – Vahnundasis 138.0 kV #1: 107.7% loading (254 MVA) following the outage of the Kensington – Vahnundasis 138.0 kV #2; DFAX = 10.3%.

For the loss of Kensington – Vahnundasis 138.0 kV Cir. 1, the Kensington – Vahnundasis 138.0 kV Circuit 2 loads to 107.7% of its 254 MVA rating. Project DFAX: 10.3%.

For the loss of Vahnundasis – Van Ness 138.0 kV, the Carrollton-Josey Lane – Sobieski 138.0 kV loads to 106.2% of its 192 MVA rating. Project DFAX: 10.4%.

For the loss of Caokia – Grant Avenue 138.0 kV Line, the Van Ness – Vahnundasis 138.0 kV loads to 101.3% of its 232 MVA rating. Project DFAX: 9.7%.

The Colesville – Darien Substation 138.0 kV is overloaded to 100.3% of its 317 MVA emergency rating for the loss of the Adrian Energy Associates Llc 345 kV/138 kV Transformer. The project contributes 12.0% DFAX (53.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Seek Lane – Bear Swamp 138.0 kV	Rebuild line with higher-capacity conductor	\$3,903,000	100% (\$3,903,000)
Bear Swamp – Wheaton 138.0 kV	Rebuild line with higher-capacity conductor	\$1,980,000	100% (\$1,980,000)
Sony – Wheaton 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,494,000	100% (\$1,494,000)
Sony – Woodenshoe 138.0 kV	Rebuild line with higher-capacity conductor	\$1,136,000	100% (\$1,136,000)
Sligo – Linden 138.0 kV	Rebuild line with higher-capacity conductor	\$949,000	100% (\$949,000)
Kensington – Woodenshoe 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,719,000	100% (\$5,719,000)
Tokoma – Seek Lane 138.0 kV	Rebuild line with higher-capacity conductor	\$2,588,000	60% (\$1,554,000)
Darien – Sligo 138.0 kV line	Rebuild line with higher-capacity conductor	\$7,265,000	100% (\$7,265,000)
Grant Avenue – Seek Lane 138.0 kV	Rebuild line with higher-capacity conductor	\$1,661,000	100% (\$1,661,000)
Vahnundasis – Kensington 138.0 kV Line 1	Reconductor line	\$2,135,000	100% (\$2,135,000)
Kensington – Vahnundasis 138.0 kV Circuit 2	Reconductor line	\$2,739,000	100% (\$2,739,000)
Carrollton-Josey Lane – Sobieski 138.0 kV	Reconductor line	\$260,000	100% (\$260,000)
Vahnundasis – Van Ness Substation 138.0 kV L	Reconductor line	\$583,000	100% (\$583,000)
Colesville – Darien 138.0 kV	Reconductor line	\$400,000	30% (\$121,000)
	Total allocated to project		\$31,499,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.